

Followup 26-11

Chest decompression in a head-on collision

Patient	33-year-old male, no past medical history. Restrained driver with airbag deployment in a head-on collision at highway speed in a 70 mph zone. There was a fatality in the second vehicle. Extricated by fire before EMS arrival and placed on the pavement with police maintaining cervical spine control.
Outcome	Survived. Severe traumatic brain injury with grade 2 diffuse axonal injury, left diaphragmatic rupture with abdominal contents herniated into the chest , left hemopneumothorax, left ribs 3 through 11 with three of them fractured in two places, right ribs 2, 4, 6 and 7, a sternal fracture, an open-book pelvis with complex sacral fracture and retroperitoneal hematoma, a right open ankle fracture, and an open great toe fracture.
Disposition	Eleven days of operative management across five trips to surgery, including splenectomy and diaphragm repair, colostomy, ankle fixation, and lumbopelvic fusion. Tracheostomy on day 7. Discharged to a comprehensive rehabilitation program, following single-step commands, with his family present.

The call

Four units worked this incident: one BLS ambulance, two Echo units, and an ALS ambulance that arrived late staffed by administrative personnel. The times below come from the Echo record; the BLS clock runs about three minutes ahead of it.

On arrival he was on the pavement beside the vehicle, extricated by fire, clothing cut away, cervical collar in place, police holding his head. The scene account was a head-on collision at highway speed.

The two crews described his mental status differently and both were right. The BLS narrative records him unconscious, with no eye opening to verbal or painful stimulus, thrashing at random intervals without becoming alert. The Echo narrative records him altered but responding to pain, combative due to confusion, moving all extremities. Those accounts are minutes apart, and what they describe is a level of consciousness that was moving.

Course, relative to Echo patient contact:

From contact	Finding or action
+0	Airway, breathing, circulation. Neurologic assessment. Full assessment.
+1	Chest deformity with left-sided crepitus , lung sounds present, chest movement present. Open right ankle fracture. Abdomen soft, pelvis intact. Multiple abrasions. Very weak radial pulses, capillary refill about 3 seconds.
+1	Pulse 140, respirations 20, SpO2 90% , glucose 140. Blood pressure unobtainable.
+2	Sam splint applied to the ankle, kicked off during loading.

From contact	Finding or action
+3	IV access.
+6	Rigid Velcro splint applied, stayed in place until hospital staff removed it.
+6	Nasopharyngeal airway.
+7	Oxygen 15 L by nasal cannula.
+9	Respirations 23, SpO2 88%.
+9	Fentanyl 100 mcg IV.
+11	Pulse 140, respirations 20, SpO2 82%. Blood pressure unobtainable.
+12	Respirations 35, SpO2 81%.
+12	Cardiac monitor: sinus tachycardia 140 to 150. Leads would not stay on because of movement and diaphoresis.
+13	Tranexamic acid 1 gram IV.
+15	Respirations 32, SpO2 85%.
+18	Oxygen changed to non-rebreather at 15 L.
+19	Secondary survey: breath sounds now absent on the left. Chest decompression performed.
+19	Pulse 150, respirations 29, SpO2 93%. Chest movement improved.
+23	Respirations 15.

An intraosseous line was attempted and abandoned because he would not tolerate it. Blood pressure was never obtained at any point during the call.

The BLS crew ran a parallel record on the ankle: open fracture identified after cutting the boot away, bleeding controlled with gauze and a pressure dressing, distal pulse found and **marked on the skin in ink**, and circulation, motor and sensation checked and documented before and after each of the two splints.

The three things to hold onto are the saturation trend, the moment the breath sounds changed, and the fact that a blood pressure was never successfully obtained.

Primary impression

Multisystem trauma with chest injury and a tension pneumothorax.

The chest call was right, the decompression was right, and the patient improved after it. What was actually in that hemithorax was not what was anticipated.

The receiving hospital found a **large left diaphragmatic rupture with abdominal contents herniated up into the mid chest, displacing the mediastinal structures including the heart to the right**. They also found a left hemothorax on FAST, a left lung contusion, multiple left rib fractures, and subcutaneous gas in the left chest wall.

So the pressure in that chest was stomach and bowel and blood, with air in the mix. The physical findings were identical to a tension pneumothorax, the physiology was identical, and the decompression was followed by better chest movement and a saturation that went from 81% to 93%.

That is not a miss. Nothing available in the field distinguishes those two things, and the presentation that prompted the needle is the presentation that should prompt it. The value in knowing the difference is for cases where needle decompressions don't yield the same result.

Hospital course

First hospital. He arrived in a collar after field decompression, tachycardic and hypotensive, moaning with his eyes closed, moving all four extremities spontaneously, **GCS 7 to 8**. A left chest tube was placed and he was intubated. A tibial intraosseous line was established and whole blood was started. FAST identified a left hemothorax. Chest radiograph showed the diaphragmatic rupture with abdominal contents in the mid chest and the mediastinum pushed right, plus lung contusion, multiple left rib fractures, and subcutaneous gas **without pneumothorax**.

His blood pressure stabilized enough for CT, which showed a **widened pubic symphysis and severe sacral fractures with a hematoma and signs of active bleeding**. A pelvic binder went on and massive transfusion protocol was activated. He received cefazolin, tetanus vaccine, and tranexamic acid, and was transferred to a second hospital for operative management.

Receiving hospital, days 0 through 14. GCS 3 at the referring hospital, 7T on arrival, in ongoing hemorrhagic shock. Initial management was massive transfusion, pelvic binder, chest tube, and central venous access, with interventional radiology angiography of the pelvis finding no active bleeding by that point.

The operative sequence:

- **Day 1.** Exploratory laparotomy with splenectomy, left diaphragm repair, partial sigmoidectomy, and end colostomy.
- **Day 2.** Irrigation and debridement with internal fixation of the right ankle, and an anterior pelvic external fixator.
- **Day 4.** Open fixation of the anterior pelvic ring with plate and screws.
- **Day 5.** External fixator removed to allow prone positioning, then closed reduction of a sacral dislocation, open reduction and internal fixation of the sacral fracture, and lumbosacral pelvic fusion with instrumentation and bone grafting.
- **Day 11.** Percutaneous posterior pelvic ring fixation with transiliac transsacral screws.

He remained intubated and sedated until **day 7**, when a percutaneous tracheostomy was performed. MRI of the brain the same day confirmed **grade 2 diffuse axonal injury involving both cerebral hemispheres and the corpus callosum**.

He was weaned to a tracheostomy collar, began following single-step instructions, and was discharged to a comprehensive rehabilitation program with his partner and mother present and supportive.

Final diagnoses: severe traumatic brain injury with grade 2 diffuse axonal injury and scattered intraparenchymal hemorrhages; left diaphragmatic rupture; left hemopneumothorax; multiple bilateral rib fractures and sternal fracture; open-book pelvis with complex lumbosacral fracture and retroperitoneal hematoma; right open Maisonneuve variant ankle fracture; open left great toe fracture; hemorrhagic shock.

Education points

What was actually happening in the thorax

A tension pneumothorax and a ruptured diaphragm produce the same picture, and understanding why makes both of them easier to recognize.

Start with the pressure problem. The chest is a closed box holding two lungs, the heart, and the great vessels. Anything that takes up space in one side of it has to displace something. In a tension pneumothorax, air enters the pleural space through a one-way defect and cannot leave, so the pressure rises with every breath. The lung on that side collapses, and then the mediastinum, meaning the heart and great vessels, gets pushed toward the other side. **The reason that kills is not the collapsed lung, it is the obstruction.** The vena cava is a floppy, low-pressure tube, and shifting and twisting it obstructs the blood returning to the heart. Without venous return, preload is nonexistent; there is nothing to pump, which is why a tension pneumothorax presents as shock.

Now put something else in the same space. The diaphragm is a sheet of muscle separating chest from abdomen. A sudden, violent increase in abdominal pressure, which is exactly what a head-on collision with a lap belt does, can tear it. When it tears, the pressure gradient across it does the rest: the abdomen normally sits at a higher pressure than the chest, so stomach, bowel, spleen and omentum are pushed upward through the defect into the hemithorax.

The result is identical. Something occupying the chest, a lung that cannot expand, a mediastinum pushed across, and obstructed venous return. **Absent breath sounds on that side, hypoxia, tachycardia, shock.**

Ninety percent of these are on the left, and the reason is anatomical. The liver sits directly under the right hemidiaphragm and acts as a plug, spreading the force and blocking herniation. The left side has only stomach and bowel beneath it, which offer no such protection. His was left.

Two important thoracic lessons from this case. The first is that the correct action is the same regardless of the actual pathology. A patient with absent unilateral breath sounds, falling saturation, and shock after high-energy blunt trauma buys a needle, because tension pneumothorax is the one you can treat in the field and nothing available to you tells the causes apart.

The patient in this case improved because there genuinely was air in that chest as well. The receiving hospital documented a left hemopneumothorax and subcutaneous gas in the chest wall, alongside the herniated contents and the blood. The needle relieved the part of the problem it could relieve.

The second is what to think when it does not work. Start with the catheter. Roughly a third of needle decompressions fail to reach the pleural space at all, which makes it the most common reason a decompression does nothing. Chest wall thickness is why, and it defeats a standard 14 gauge catheter at four and a half to five centimeters in obese or heavily muscled patients. Needle length matters more than anything else you control, with failure rates falling by about eight percent for every additional centimeter.

Site matters too, and it is not one answer. The anterior axillary approach crosses a thinner chest wall, which is the argument for it and the reason ATLS moved there in 2018. A 2025 meta-analysis in the World Journal of Emergency Surgery, pooling 51 studies, adds a side-specific caveat: it favors the second intercostal space at the midclavicular line for left-sided decompression, on the grounds that the lateral sites carry a higher risk of cardiac injury on that side. The authors qualify it themselves, citing heterogeneity between studies and variability in measurement, and no guideline currently makes that distinction. Worth being aware of, but not yet settled.

If the needle reached and the patient did not improve, something other than air is filling the hemithorax. The common reason is a massive hemothorax, which is far more frequent than a diaphragmatic rupture, and blood

will not come out through a 14 or 12 gauge catheter. Rarely it is a diaphragmatic rupture with abdominal contents in the chest, which is what this patient had alongside the blood and air.

When a decompression does not have the desired effect, work it in that order: did the needle reach, is the hemithorax full of blood, and then the uncommon causes. What does not change is the destination or the urgency. All of them need a chest tube and a surgeon, and none of them is fixed definitively on the roadside

Further reading on decompression site and needle length. The figures above come from a 2025 meta-analysis in the World Journal of Emergency Surgery, free to read in full: <https://pubmed.ncbi.nlm.nih.gov/40383767/>

A shorter summary of the site debate, including the ATLS and European Trauma Course split, is at <https://hospitalprocedures.org/site-of-needle-decompression/> It is a commercial training site, so expect the course advertising alongside it.

The decision to decompress

The route to that decompression is worth walking through, because it was not a classic presentation and nobody was handed the answer.

At the first assessment, one minute in, he had **chest deformity and crepitus on the left, and lung sounds were present**. A tension pneumothorax that is not yet tensioning does not have absent breath sounds. What it has is a mechanism, a chest wall injury, and time.

Then the numbers moved. Saturation 90, then 88, then 82, then 81, while the respiratory rate climbed 20, 23, 20, 35. **He was working harder and achieving less**, and this happened across two escalations of oxygen: a nasal cannula at 15 liters, then a non-rebreather at 15 liters. Neither helped, because the problem was never the oxygen concentration. It was that one lung was not participating.

On the secondary survey the breath sounds on the left were gone. That is the finding that made the diagnosis, and it was available only because somebody examined the chest a second time.

Tension is a process, not a state. The patient you assess at minute one and the patient you have at minute fifteen are not the same patient. A chest injury with sounds present is a chest injury that needs to be checked again.

Agitation, and its costs

He was thrashing from the moment the first crew reached him, and the record documents exactly what that cost: **a blood pressure, the monitor leads, an intraosseous line, and the first splint**. At no point in the entire call did anyone successfully obtain a blood pressure.

The important reframing is that **the agitation was the presentation, not an obstacle to it**. A patient with a severe traumatic brain injury, a saturation in the eighties, and shock from obstructed venous return is agitated for those reasons. The thrashing was the clearest live signal of how his brain and his oxygen delivery were doing, and it was the same finding both crews documented in different words.

The two accounts also demonstrate something about head injury that is easy to miss. A level of consciousness that swings between no response to pain and confused resistance within a few minutes is **labile**, and lability is itself a finding. It belongs in a handoff. When two competent people describe the same patient differently minutes apart, the variation is data rather than a disagreement to resolve.

Diffuse axonal injury

His brain injury was confirmed on MRI on day 7 as **grade 2 diffuse axonal injury involving both cerebral hemispheres and the corpus callosum**. It is worth understanding, because it is the brain injury that a crew sees most often in a high-speed collision and the one that CT imaging is worst at showing.

Brain tissue is not uniform. Grey matter and white matter have different densities, so when a head is accelerated or decelerated violently they do not move at quite the same rate. The mismatch is worst where the two meet, and worse again with rotational force than with a straight-line impact, because rotation makes different parts of the brain travel different distances in the same instant.

What tears is the axons. **An axon is the long projection a neuron sends out to reach other neurons**, some of them running the full width of the brain, and white matter is essentially bundles of them. They tolerate stretch badly. A rapid shearing force pulls them past what the internal scaffolding can absorb.

The tearing is usually not immediate. The stretch damages the structure inside the axon, transport along it fails, the contents pile up at the injured point, and the axon swells and disconnects over the following hours to days. **That is why it is called diffuse axonal injury rather than axonal tearing**, and it is why a patient can be deeply unconscious from the moment of impact with a CT that looks close to normal. The damage is microscopic and spread across millions of connections rather than concentrated in one bleed.

Grading follows how deep the shearing reaches. Grade 1 stays at the grey and white matter junction. **Grade 2, which he had, adds the corpus callosum**, the thick bundle joining the two hemispheres. Grade 3 reaches the brainstem and carries the worst outlook.

Two important considerations. **CT is poor at finding it**, which is why his diagnosis came from an MRI a week later rather than from the scan on arrival, and why his initial imaging showed only scattered small hemorrhages. And **the mechanism is rotational deceleration**, so it needs no skull fracture and no large bleed at all. A patient can have a devastating brain injury and a scan that looks almost reassuring.

The clinical picture is the one both crews documented and could not reconcile: **unconsciousness out of proportion to anything you can see**, with a level of consciousness that moves. He was GCS 3 at the first hospital and 7T on arrival at the second, and he was following single-step instructions by discharge. That trajectory is normal for grade 2 and it is measured in months.

The pelvis missed until pan-scan

The field record says the pelvis appeared intact. CT later found a **widened pubic symphysis and severe sacral fractures with a hematoma and signs of active bleeding**, and he went on to have an external fixator, anterior ring plating, sacral reduction and fixation, lumbopelvic fusion, and transiliac transsacral screws across four separate operations.

Two experienced paramedics palpated that pelvis and felt nothing. The emergency department missed it on both the primary and the secondary survey. It was found on imaging.

That is worth knowing because it is the expected performance of the test rather than a lapse. Physical examination of the pelvis is unreliable, and it gets worse in exactly the patients who need it most. An alert patient who can tell you it hurts gives you useful information. An obtunded, distracted, agitated patient with a severe brain injury and an open ankle fracture pulling all of his attention gives you almost nothing, because the exam depends on a pain response he cannot produce reliably. Add the fact that an open-book injury can be reduced back into near-normal position by lying supine on a hard surface, and a fracture that is actively bleeding can feel entirely stable under your hands.

There is a second reason not to lean on it. **Repeatedly springing or rocking a pelvis to test for instability can disrupt clot that has already formed**, and the modern teaching has moved away from it for that reason. You are performing a test with poor sensitivity at the cost of potentially worsening the bleeding you are testing for.

Prehospital considerations

Trauma triage was met on multiple red criteria. Berrien County MCA protocol 2-1 lists, among the red criteria, chest wall instability or deformity, inability to follow commands with a motor GCS below 6, a respiratory rate above 29, room air pulse oximetry below 90%, and a heart rate greater than the systolic pressure. He met several of them within the first minute, which makes destination a settled question rather than a judgment call.

Protocol 2-2 (General Trauma) sets the decompression threshold at suspicion. It directs high-flow oxygen for chest injury, says to avoid positive pressure ventilation if possible, and states that if tension pneumothorax is suspected, perform needle decompression. You do not have to be certain.

No blood pressure was ever obtained on this patient, and it was not for lack of trying. Automated and manual attempts both failed, and two separate things were working against them. He was thrashing continuously, and neither an oscillometric cuff nor an ear on a stethoscope can find Korotkoff sounds on a limb that will not stay still. And he was poorly perfused, with very weak radial pulses and a capillary refill around three seconds, which is exactly the state in which both methods stop producing a number. Auscultation fails when the pulse pressure is too narrow to generate audible sounds, and the automated cuff fails for the same reason through a different sensor. **The physiology that made the reading important is the physiology that made it unobtainable.**

What the crew used instead is the answer. Weak radial pulses, capillary refill, skin findings, and a heart rate of 140 to 150, all recorded, with the cuff charted as unobtainable rather than left blank. **In a patient like this, the surrogates are the vital sign.**

The interventions

PREHOSPITAL INTERVENTIONS

Chest decompression. The intervention that mattered, performed on absent unilateral breath sounds with a falling saturation in a shocked trauma patient, which is the indication. Chest movement improved and saturation went from 81% to 93% immediately afterward, and the respiratory rate fell to 15 within four minutes. The receiving hospital placed a chest tube on arrival, which is the definitive version of the same procedure.

Oxygen, escalated twice. Nasal cannula at 15 liters, then non-rebreather at 15 liters. Worth a note: a nasal cannula is not designed to run at 15 liters and does not deliver a high fraction of oxygen at any flow, because the patient still entrains room air around it. The non-rebreather was the right device. Neither fixed him, and that failure to respond is part of what pointed at the chest.

Nasopharyngeal airway. Appropriate in a patient with a depressed level of consciousness who is still breathing and would not tolerate anything in his mouth.

Splinting, with a documented circulation chain. This is the best-documented BLS work this program has seen. The boot was cut away, the open fracture identified, bleeding controlled with gauze and a pressure dressing, the distal pulse located and **marked in ink**, a sam splint applied and then kicked off, and a rigid Velcro splint applied that held until the hospital removed it. Circulation, motor and sensation were checked and documented **before and after each splint**, and the findings were passed to the paramedic before departure.

Protocol 2-5 asks for exactly that: evaluate and document neurovascular status before and after splinting. Marking the pulse in ink is not in any protocol and it is the detail that makes the rest of it useful, because it lets the next person find the same spot.

Tranexamic acid 1 gram IV at 13 minutes. Well inside the window, in a patient with a high-energy mechanism, an open fracture, and shock physiology. Straightforwardly correct.

Fentanyl 100 mcg IV. An open ankle fracture in a patient who is thrashing is a real indication, and a splint that will not stay on a moving limb is worse than no splint. Nobody should be criticized for treating that.

The timing is worth talking through, because this patient had no measurable blood pressure, a heart rate of 140 to 150, very weak radial pulses, and a saturation of 88 falling toward 81. Ten minutes later he was decompressed for tension physiology that had been building since first contact.

Fentanyl does three things in that patient. It reduces sympathetic outflow, which is the tone holding a blood pressure up in someone who is compensating. It depresses respiratory drive, which is part of what was keeping his saturation where it was. And it removes agitation, which in this patient was the clearest ongoing sign of how his brain was doing.

Ketamine is worth considering here, and the reason is the movement rather than the pain. His thrashing cost the crew a blood pressure, the monitor leads, an intraosseous line, and a splint. That is an access problem rather than a comfort problem, and it was obstructing the assessment that eventually found the chest. Ketamine dissociates without dropping sympathetic tone the way an opioid does, and it preserves respiratory drive and airway reflexes. The tradeoff is the same one in a different form: sedating him also removes the agitation you were using as a neurologic monitor. Note that the dose required here is dissociative, not the analgesic dose in the pain management protocol, and reaching it for this indication needs a medical control order.

Nothing bad demonstrably followed the fentanyl. His saturation was already falling before the dose and kept falling after it, which is the chest rather than the drug, and his respiratory rate went up rather than down. But **the agitation that made him hard to treat was also the finding signaling how sick he was**, and any sedation in an undifferentiated shocked trauma patient spends some of the reserve he is running on. Whichever drug, give it in increments and watch what it costs.

Antibiotics were not given, and that was the right call. Protocol 2-5 directs antibiotics for open fractures, and Berrien has selected cefazolin 2 grams by slow intravenous push. It is a real intervention with real evidence behind it. It also does not compete with tension physiology, an unmeasurable blood pressure, and a saturation of 81. The splint was on and staying on after two attempts, the limb had documented circulation, motor and sensation, and disturbing that to give a drug whose benefit is measured in infection rates weeks later would have been the wrong trade. **The thing that would have killed him got the attention. The thing that would have cost him a wound infection waited**, and the receiving hospital gave cefazolin on arrival.

IN-HOSPITAL INTERVENTIONS

Chest tube. The definitive version of what the needle started. A needle relieves pressure through a small catheter that blocks easily and does not drain blood. A chest tube is a large-bore tube placed under direct vision that evacuates air and blood continuously and stays in place. In a patient with a hemothorax as well as air, the drainage is the point.

Intubation. For a GCS of 7 to 8 with a chest that was not working and an operative course ahead of him.

Intraosseous access and whole blood. The intraosseous line the field crew could not place was established in the resuscitation bay on a sedated patient, and whole blood was started through it. Whole blood gives red cells,

plasma and platelets in the proportions they were lost in, rather than reconstituting them from separate components.

FAST examination and chest radiograph. The FAST found the left hemothorax. The chest film found the diaphragmatic rupture, and it is worth noting how: **a plain radiograph showed abdominal contents in the mid chest and the heart displaced to the right.** That diagnosis was available in under a minute from a bedside film.

CT of the chest, abdomen and pelvis, which found the widened pubic symphysis, the severe sacral fractures, and signs of active bleeding.

Pelvic binder and massive transfusion protocol. An open-book pelvis increases the volume of the pelvic ring and lets it bleed without limit. A binder closes the ring mechanically, reducing that volume and allowing tamponade. Massive transfusion protocol delivers red cells, plasma and platelets in fixed ratios rather than crystalloid, which is what a patient losing whole blood actually needs.

Interventional radiology angiography, looking for an arterial bleeder in the pelvis to embolize. It found none, which means the bleeding was venous and from bone surfaces, and the binder plus transfusion was the treatment.

The operative sequence, and the reason behind the timing. Day one was damage control: stop the bleeding, repair the diaphragm, deal with the injured bowel and spleen, and get out. The pelvis got an external fixator on day two, which is a temporary frame that stabilizes the ring quickly without a long operation. Definitive internal fixation came on days four, five and eleven, once he was stable enough to tolerate hours of surgery and prone positioning. **That staging is deliberate.** Definitive repair of everything on day one would have killed him.

Tracheostomy on day 7. After about a week of intubation, a tracheostomy reduces sedation requirements, protects the larynx from pressure injury, and makes weaning easier in a patient who will need a ventilator for weeks.

The labs, and what they mean

Reference ranges vary between institutions and analyzers. These are the ranges printed on the source reports.

Lab	Value	Reference range
Prothrombin time (seconds)	12.8, then 13.3	9.5 to 12.0
INR	1.3	0.9 to 1.2
Fibrinogen (mg/dL)	166, then 135, then 164	190 to 450
Ionized calcium (mmol/L)	0.92, then 0.93, then 0.97	1.12 to 1.40
White blood cell (x10 ⁹ /L)	15.3, then 20.5, then 17.5	4.0 to 10.8
Hemoglobin (g/dL)	13.9	14.0 to 18.0
Hematocrit (%)	41.8	42.0 to 52.0

These numbers are trauma-induced coagulopathy, and they arrived early.

A prothrombin time and INR that are mildly prolonged in a young man with no medical history and no anticoagulants are not a laboratory error. Severe injury switches on the clotting cascade throughout the body, consumes clotting factors faster than the liver can replace them, and simultaneously activates the system that

dissolves clot. **His fibrinogen tells that story most clearly.** Fibrinogen is the protein clot is built from, and it is an acute phase reactant, which means it should rise when a patient is acutely injured. His fell to 135 against a normal floor of 190. That is consumption outrunning production.

The ionized calcium is the one worth memorizing. His was 0.92, well below a normal floor of 1.12, and it stayed low across three draws. Calcium is required for the clotting cascade, where it is designated factor IV, and it is also required for myocardial contractility and vascular smooth muscle tone. A bleeding patient with low ionized calcium clots worse, pumps worse, and holds their blood pressure worse, and all three make the bleeding worse.

Two things drive it down. Severe injury itself lowers it before any treatment, and **stored blood products are anticoagulated with citrate, which binds calcium.** A patient receiving massive transfusion is being given a calcium binder with every unit. This is why calcium replacement is part of massive transfusion protocols and why it sits alongside acidosis, hypothermia and coagulopathy in how trauma teams think about a patient who will not stop bleeding.

His hemoglobin of 13.9 is a trap. It is essentially normal, in a man with a hemothorax, a retroperitoneal hematoma, a bleeding pelvis, and an open fracture. Hemoglobin is a concentration, not a volume. Whole blood loss removes red cells and plasma in the same proportion, so the concentration does not fall until interstitial fluid shifts in or crystalloid is given, and that takes hours. **An early normal hemoglobin in an actively bleeding patient tells you almost nothing,** and his fell steadily once resuscitation began.

The white count of 15.3 rising to 20.5 is the reactive leukocytosis of major injury rather than infection.

Learning points

1. **A tension pneumothorax and a ruptured diaphragm present identically, and the field action is the same.** Both put something in a hemithorax that pushes the mediastinum across and obstructs venous return, which is why both present as absent breath sounds with hypoxia and shock. Ninety percent of diaphragmatic ruptures are on the left, because the liver plugs the right. Needle a patient with that presentation. If they do not improve, you have not missed the spot, you have learned what is in there.
2. **Tension is a process, so the exam has to be repeated.** At one minute this patient had chest deformity and crepitus with breath sounds present. At nineteen minutes the sounds on that side were gone. Nothing changed except time, and the only reason anybody found it was a second look at the chest. A saturation that falls through two escalations of oxygen is telling you the problem is not oxygen concentration.
3. **Agitation in a shocked head-injured patient is the presentation, not an obstacle to it.** His thrashing cost a blood pressure, the monitor leads, an intraosseous line, and a splint. It was also the clearest live signal of his brain and his oxygen delivery. Two crews described him differently minutes apart, and that lability is a finding worth handing over rather than a disagreement to settle.
4. **When the cuff will not read, the surrogates become the vital sign.** Automated and manual attempts both failed on this patient, because he was thrashing and poorly perfused, and those two things defeat both methods. The physiology that made the number important is what made it unavailable. Weak radial pulses, a capillary refill around three seconds, and a heart rate of 140 to 150 are what this crew used and recorded, and they carry the information the cuff would have.
5. **When two correct interventions compete, the one that kills first wins.** Antibiotics for an open fracture are indicated and Berrien carries cefazolin for it. They also do not compete with a chest that is not working and a patient with no measurable pressure. The splint was on and holding, the limb had documented circulation,

and the wound could wait for the hospital. Triage between good options is a decision, and it deserves to be documented as one.

Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at <https://bcmca.github.io/followup/>